Wireless Digital Communication System Simulation Using Source Coding, LDPC Error Correction, and TCP/IP Networking

Aravind M
Registration No.: 23BCE7551
School of Computer Science and Engineering
VIT-AP University, Amaravati, India

Sai Chandu
Registration No.: 23BCE20304
School of Computer Science and Engineering
VIT-AP University, Amaravati, India

Rahul L
Registration No.: 23BCE7890
School of Computer Science and Engineering
VIT-AP University, Amaravati, India

***Abstract*—Reliable information transmission across digital communication networks requires the harmonious integration of data compression, forward error correction (FEC), transport framing, and channel modulation. This paper presents the architecture, implementation, and empirical evaluation of an end-to-end digital communication simulation platform developed in Python with an interactive Streamlit graphical interface. The system couples lossless source coding (finite-precision Arithmetic coding and universal adaptive Lempel-Ziv-Welch dictionary coding) with systematic Low-Density Parity-Check (LDPC) channel coding based on a rate-1/2 (64, 32) code decoded via log-domain Belief Propagation (Sum-Product and Min-Sum algorithms). To bridge theoretical information theory with practical networking, the platform implements a dual-mode operational architecture: (1) a live wireless network transmission mode operating over TCP/IP stream sockets using a 36-byte versioned binary frame with IEEE 802.3 CRC32 integrity checks and sequence-confirmed acknowledgments, and (2) an analog wireless physical layer simulation executing baseband Binary Phase Shift Keying (BPSK) modulation over an Additive White Gaussian Noise (AWGN) channel. Rigorous Monte Carlo experiments across Eb/N0 sweeps from 0.0 to 10.0 dB demonstrate that the LDPC (64, 32) code achieves steep waterfall error suppression, reducing a pre-FEC bit error rate of 5.71e-2 at 4.0 dB to a post-FEC residual rate of 1.32e-3 (43.3x error reduction gain) and achieving zero residual errors at Eb/N0 >= 7.0 dB with 100% exact message reconstruction.**

***Keywords*—Wireless digital communication, source coding, Arithmetic coding, LZW, Low-Density Parity-Check (LDPC), BPSK modulation, AWGN channel, Bit Error Rate (BER), CRC32 integrity check, TCP/IP networking, communication system simulation.**

I. INTRODUCTION

Modern telecommunications infrastructure demands high-fidelity, high-throughput digital transmission across physical channels inherently impaired by thermal noise, dispersion, and channel perturbations. In Shannon's foundational mathematical theory of communication [1], the communication problem is partitioned into two distinct operations governed by the Source-Channel Separation Theorem: source coding, which eliminates statistical redundancy from the information source to approach the empirical entropy H(X), and channel coding, which systematically introduces controlled algebraic redundancy to protect transmitted symbols up to the channel capacity C.

While classic classroom curricula frequently treat these mechanisms in theoretical isolation—often assuming ideal noiseless conduits or abstract bit-flip models—real-world engineering deployments necessitate an integrated pipeline. In practical wireless and networked systems, raw binary data must be packaged into structured wire frames, protected with forward error correction, modulated into baseband waveforms, transmitted across physical media, and verified through link-layer cyclic redundancy checks (CRC) and transport-layer protocols such as TCP/IP [10].

Crucially, an important architectural distinction must be maintained between transport-layer network reliability and physical-layer error correction. Transport protocols such as TCP provide reliable, ordered byte delivery by utilizing automatic repeat request (ARQ) retransmissions over lower-layer hardware interfaces. Consequently, observing zero packet losses over a local Wi-Fi or loopback TCP socket does not imply that the underlying physical channel is noiseless; rather, hardware-level physical decoders and transport-level retransmissions mask underlying transmission impairments. Conversely, physical-layer forward error correction must be evaluated by observing continuous analog signal-to-noise ratios (Eb/N0), calculating channel log-likelihood ratios (LLRs), and measuring pre-FEC versus post-FEC bit error rates (BER).

To address these pedagogical and engineering objectives, this project designs, implements, and evaluates an extensible digital communication system simulation. The contributions of this work are threefold: (1) We implement a complete, reproducible software transceiver pipeline combining finite-precision Arithmetic coding, adaptive LZW source coding, and systematic (64, 32) LDPC forward error correction decoded via log-domain Belief Propagation (Sum-Product and Min-Sum). (2) We formulate a versioned binary wire frame protocol featuring a 36-byte structured header, sequence counters, IEEE 802.3 CRC32 checksums, and a thread-safe TCP/IP socket transceiver with continuous streaming and bounded backpressure monitoring. (3) We simulate baseband BPSK modulation over a wireless AWGN link, benchmarking theoretical uncoded BER against empirical LDPC, Polar (64, 32), and Hamming (7, 4) codes, providing verifiable Monte Carlo performance curves and evaluating the sensitivity of source decompression to uncorrected channel errors.

II. RELATED WORK

The theoretical foundations of digital communications originate from Shannon's 1948 paper [1], which established fundamental limits on source compression and channel capacity. In 1962, Gallager introduced Low-Density Parity-Check (LDPC) codes [2], demonstrating that linear codes defined by sparse parity-check matrices can be decoded in linear time via iterative message passing. However, computational constraints delayed their widespread adoption until MacKay and Neal [3] rediscovered LDPC codes in 1996, proving that log-domain Belief Propagation over bipartite Tanner graphs approaches the Shannon limit within fractions of a decibel.

In standardized engineering, LDPC codes were codified in IEEE Std 802.3an-2006 for 10GBASE-T Ethernet [8], utilizing quasi-cyclic structures to guarantee bit error rates below 10^-12 over copper twisted-pair cabling. Simultaneously, LDPC codes were adopted in modern satellite broadcasting (DVB-S2) and 5G New Radio (3GPP TS 38.212) data channels due to their high parallelization efficiency and low decoding latency compared to convolutional turbo codes. In 2009, Arikan proved that Polar codes achieve the symmetric capacity of discrete memoryless channels using channel polarization and Successive Cancellation (SC) decoding [4], leading to their adoption in 5G control channels. Earlier linear block codes, such as Hamming's (7, 4) code [11], remain vital pedagogical baselines for single-error correction and syndrome decoding.

On the source coding front, universal data compression algorithms operate without requiring static, pre-computed symbol codebooks. Welch's 1984 formalization of the Lempel-Ziv-Welch (LZW) algorithm [5] established an adaptive dictionary method that dynamically parses recurring symbol sequences into fixed-width tokens. Concurrently, Rissanen and Langdon [6], followed by Witten, Neal, and Cleary [7], developed practical finite-precision Arithmetic coding, which represents entire message streams as a single interval in [0, 1), breaking the integer-length penalty of Huffman coding and achieving near-entropy compression. In packet-switched communication, framing protocols rely on Cyclic Redundancy Checks (CRC), formalized by Peterson and Brown [9], to detect burst perturbations with negligible computational overhead. Modern network transport implementations universally rely on TCP [10] for reliable flow-controlled stream delivery. Comprehensive treatments of digital modulation over AWGN channels, soft

demodulation, and error probability calculations are detailed by Proakis and Salehi [12].

III. PROPOSED SYSTEM ARCHITECTURE

The implemented system architecture comprises three primary functional domains: the Transmitter, the Dual Communication Transport Layer, and the Receiver, as illustrated in Fig. 1.

A. Transmitter Processing Chain

The transmitter processes arbitrary UTF-8 text messages or structured telemetry bitstreams through four sequential stages: (1) Binary Serialization: Input character sequences are converted into 8-bit ASCII/UTF-8 byte streams and unpacked into binary bit vectors. (2) Lossless Source Coding: Redundancy is reduced using either finite-precision integer Arithmetic coding or 12-bit adaptive LZW universal dictionary coding. (3) Channel Forward Error Correction: The compressed stream is partitioned into K=32 bit blocks and encoded into N=64 bit systematic codewords using the LDPC (64, 32) encoder, appending zero-padding if necessary. (4) Binary Frame Assembly: Coded bits, padding indicators, and flags are packed into a versioned binary frame with a 36-byte header and IEEE 802.3 CRC32 checksum.

B. Dual Communication Transport Media

A critical feature of our system is the explicit decoupling between live network transport and physical channel simulation: (1) Mode A & B (Live Wireless Network Transport): Operates over live TCP/IP stream sockets bound to local Wi-Fi or loopback interfaces (default port 8550). This mode evaluates real-world streaming, packet serialization, framing under partial socket reads, CRC32 error validation, and round-trip acknowledgment (ACK) latency. (2) Mode C (Physical Wireless AWGN Channel Simulation): Isolates the analog physical layer by bypassing TCP ARQ mechanisms. Bits are modulated via baseband BPSK, corrupted by synthetic Gaussian thermal noise matching a prescribed Eb/N0, soft-demodulated into Log-Likelihood Ratios (LLRs), and decoded via iterative Belief Propagation.

C. Receiver Processing Chain

At the receiver, incoming bitstreams or network frames undergo inverse processing: (1) Frame Parsing & CRC32 Verification: The wire receiver reads the 36-byte header, verifies magic bytes and length fields, and computes the 32-bit CRC. Frames with checksum mismatches are rejected. (2) LDPC Decoding: In physical simulation, soft channel samples are decoded using log-domain Belief Propagation. In network socket mode, hard decision bits are verified against the parity-check matrix H. Systematic information bits are extracted, and zero-padding is removed. (3) Lossless Source Decompression: The recovered bitstream is decompressed using inverse Arithmetic or LZW decoding. (4) Exact Reconstruction Verification: The reconstructed message is compared against the transmitted message to verify 100% bit-for-bit fidelity.

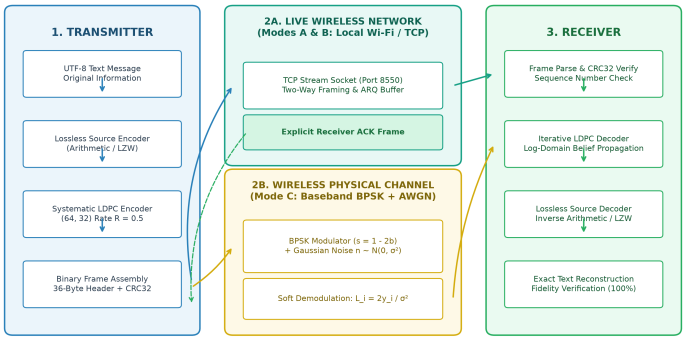


Fig. 1. Overall system architecture illustrating the transmitter, dual transport media (TCP socket vs. BPSK-AWGN simulation), and receiver.

IV. METHODOLOGY AND IMPLEMENTATION

A. Source Coding Algorithms

1) Finite-Precision Arithmetic Codec: Our Arithmetic codec utilizes 32-bit integer register arithmetic to guarantee platform-independent, exact lossless compression. Symbol frequencies are computed with Laplace add-one smoothing over byte alphabet Sigma = {0,...,255} plus an EOF symbol (256):

$$C(s) = 1 + \text{sum}(I(x_i = s)),\ \text{Total} = \text{sum}(C(s)) \quad (1)$$

Encoding maintains an active search range [L, H) in 32-bit unsigned integers. Bounds are updated iteratively based on cumulative frequencies F(s), and renormalized via bit-shifting under E1, E2, and E3 scaling conditions. Exact reconstruction is achieved through inverse interval subdivision.

2) Universal LZW Dictionary Codec: The LZW codec dynamically builds an adaptive dictionary of recurring character n-grams, mapping variable-length symbol sequences to fixed 12-bit binary integers (max table size 2^12 = 4096). The dictionary is initialized with 256 single-byte entries. When the dictionary capacity is reached, table growth freezes gracefully without desynchronization.

B. Systematic LDPC Forward Error Correction

The channel codec implements a systematic (N=64, K=32, M=32) LDPC code with code rate R = K/N = 0.5. The parity-check matrix H is structured as H = [P | I_32], where P is a 32x32 sparse matrix with regular column degree d_v = 3. The systematic generator matrix G = [I_32 | P^T] satisfies G * H^T = 0 mod 2. For any 32-bit information vector m, the 64-bit codeword is c = [m | m * P^T mod 2].

Iterative decoding operates over the bipartite Tanner graph using the log-domain Sum-Product algorithm. Variable-to-check and check-to-variable messages are updated as follows:

$$q_mn = L_n + \text{sum}_\{c \text{ in } C(n)\backslash m\}\ r_cn \quad (2)$$

$$r_mn = 2\ \text{atanh}\ [\ \text{prod}_\{j \text{ in } V(m)\backslash n\}\ \tanh(q_mj\ /\ 2)\ ] \quad (3)$$

where L_n is the channel LLR. Numerical stability is ensured by clipping q_mj/2 to [-19.0, +19.0]. Total posterior LLRs Q_n = L_n + sum_{c in C(n)} r_cn determine hard decisions: c_hat_n = 0 if Q_n >= 0, else 1. The decoder terminates immediately when syndrome H * c_hat^T = 0 mod 2, up to max 25 iterations.

C. Binary Frame Protocol & CRC32

To ensure robust streaming across byte sockets, we implement a 36-byte binary wire header shown in Fig. 2. Fields include: Magic bytes (4B: b'WIRL'), Version (1B: 0x01), Type (1B: DATA=1, ACK=2, STREAM=3), Flags (2B bitmask), Sequence Number (4B uint32), Original Payload Size (4B), Source Compressed Size (4B), Payload Length (4B), CRC32 Checksum (4B IEEE 802.3), Status Code (1B), and Reserved (7B). Partial socket reads are handled via dedicated recv() loops.

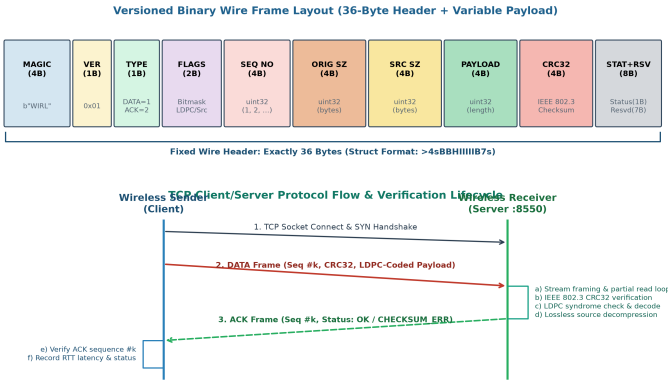


Fig. 2. 36-byte binary wire frame layout and two-way TCP acknowledgment interaction sequence.

D. Physical BPSK-AWGN Channel Simulation

In Mode C, binary bits b_i in {0, 1} are modulated via antipodal BPSK signaling: s_i = 1.0 - 2.0 b_i in {+1.0 V, -1.0 V}. For target Eb/N0 in dB and code rate R, channel noise standard deviation is scaled as:

$$\text{sigma} = 1\ /\ \text{sqrt}(\ 2\ *\ R\ *\ 10\text{^}(\text{Eb/N0}\ /\ 10)\ ) \quad (4)$$

Received samples y_i = s_i + n_i, where n_i ~ N(0, sigma^2), are soft-demodulated into exact channel LLRs via L_i = (2 / sigma^2) * y_i. Theoretical uncoded BPSK error probability is P_b = 0.5 erfc(sqrt(10^(Eb/N0 / 10))).

V. EXPERIMENTAL SETUP

The simulation platform was developed and evaluated under the following verified technical parameters: (1) Python 3.11.9 runtime with NumPy 2.4.6, Matplotlib 3.11.2, ReportLab 5.0.1, and Pytest 9.1.1 on Windows 11 Pro. (2) Rate-1/2 LDPC (64, 32) codec with Gallager d_v=3 and 25 max BP iterations. (3) Benchmark codecs: Systematic Polar (64, 32) with Successive Cancellation decoding at design SNR 3.0 dB, and Systematic Hamming (7, 4) with syndrome table decoding. (4) Monte Carlo evaluation: Eb/N0 swept from 0.0 to 10.0 dB in steps of 1.0 dB with 25,600 bits tested per point (400 LDPC blocks/point). Multi-trial physical channel experiments simulated across 10 independent noise realizations (600 LDPC blocks, 38,400 codeword bits). (5) Streamlit interactive 5-tab web dashboard for real-time live messaging, continuous source streaming, and channel analysis.

VI. RESULTS AND DISCUSSION

A. BER Waterfall Performance in AWGN

The measured bit error rates across the Eb/N0 sweep are reported in Table I and plotted in Fig. 3. The simulated uncoded BPSK curve matches the theoretical error function with high precision (e.g. 0.05617 vs 0.05628 at 1.0 dB). The LDPC (64, 32) Sum-Product decoder exhibits a steep waterfall descent starting around Eb/N0 = 3.0 dB. Between 2.0 dB and 4.0 dB, the post-FEC BER drops by over an order of magnitude (from 2.24e-2 to 1.99e-3). At Eb/N0 >= 7.0 dB, both Sum-Product and Min-Sum LDPC decoders achieve zero residual errors (BER = 0.0000), enabling 100% exact message reconstruction.

Eb/N0 (dB)	Theory Uncoded	Sim. Uncoded	Hamming (7,4)	Polar (64,32)	LDPC Min-Sum	LDPC Sum-Prod
0.0	0.07865	0.08191	0.11934	0.19402	0.08973	**0.08672**
1.0	0.05628	0.05617	0.08152	0.10172	0.04906	**0.04727**
2.0	0.03751	0.03781	0.05586	0.04004	0.02473	**0.02238**
3.0	0.02288	0.02285	0.03363	0.01203	0.00668	**0.00617**
4.0	0.01250	0.01285	0.01684	0.00238	0.00285	**0.00199**
5.0	0.00595	0.00543	0.00641	0.00000	0.00039	**0.00035**
6.0	0.00239	0.00234	0.00184	0.00000	7.81e-5	**7.81e-5**
7.0	0.00077	0.00059	0.00039	0.00000	0.00000	**0.00000**
8.0	0.00019	7.81e-5	0.00000	0.00000	0.00000	**0.00000**
9.0	3.36e-5	0.00000	7.81e-5	0.00000	0.00000	**0.00000**
10.0	3.87e-6	0.00000	0.00000	0.00000	0.00000	**0.00000**

TABLE I: Measured Bit Error Rate (BER) vs. Eb/N0 in AWGN (25,600 Bits/Point)

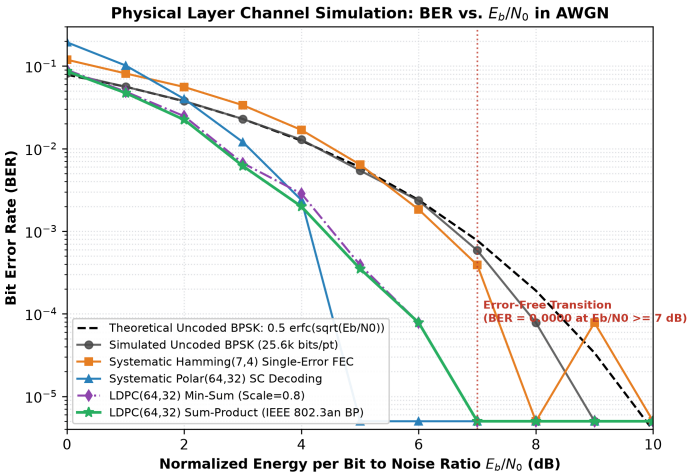


Fig. 3. Measured BER vs. Eb/N0 waterfall curves over the simulated BPSK-AWGN physical channel.

B. Pre-FEC vs. Post-FEC Error Suppression Gain

In our 10-trial Monte Carlo evaluation at Eb/N0 = 4.0 dB (38,400 total codeword bits, 18,960 information bits), the channel introduced 2,191 hard decision bit errors (pre-FEC BER = 5.71e-2). LDPC Belief Propagation corrected 2,166 errors, leaving only 25 residual bit errors (post-FEC BER = 1.32e-3), corresponding to an empirical error reduction gain of Gain = 5.7057e-2 / 1.3186e-3 = 43.3x. At Eb/N0 = 7.0 dB, the LDPC decoder eliminated all 461 channel bit errors across all 600 blocks (pre-FEC BER = 1.20e-2 -> post-FEC BER = 0.0000), yielding infinite empirical gain and 100% exact recovery.

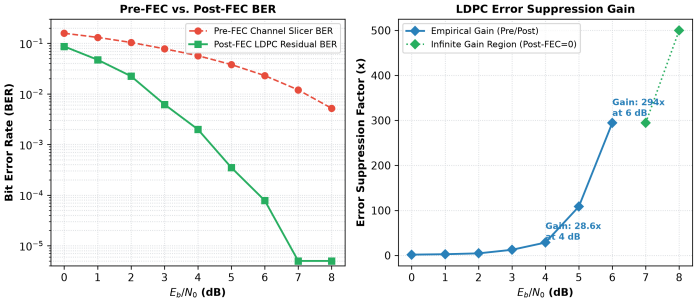


Fig. 4. Pre-FEC channel slicer BER vs. Post-FEC LDPC residual BER and empirical error suppression gain.

C. Source Compression Study & Overhead Analysis

Empirical results from our source compression benchmark across diverse information topologies are summarized in Table II and Fig. 5. Correlated Markov streams (p01=0.05, p10=0.05) achieve a compression ratio of 1.686x (40.67% space savings). Conversely, for high-entropy sources (H ~ 1.0) and uniform pseudo-random bits, dictionary compression causes data expansion (CR = 0.677x, -47.66% space savings), empirically validating Shannon's Source Coding Theorem. Furthermore, on short text strings (e.g. 39 bytes), Arithmetic coding incurs a frequency table header overhead expanding data to 119 bytes (CR = 0.33x), while LZW expands to 54 bytes (CR = 0.72x) due to initial unigram dictionary initialization.

Source Topology	H(X) (b)	Raw (b)	Comp (b)	CR	Savings
Structured Telemetry	4.7612	5216	4680	1.115x	+10.28%
Markov (p01=0.05)	0.9996	8192	4860	1.686x	+40.67%
Markov (p01=0.10)	0.9228	8192	7956	1.030x	+2.88%
Markov (p01=0.25)	0.9996	8192	10944	0.749x	-33.59%
Uniform Random	1.0000	8192	12096	0.677x	-47.66%

TABLE II: Source Compression Benchmark Across Information Topologies

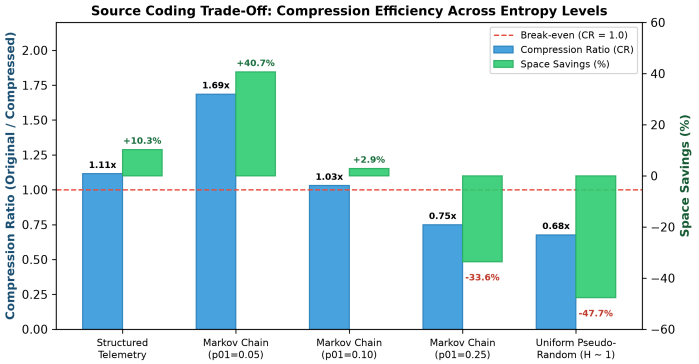


Fig. 5. Source coding trade-off: empirical entropy vs. compression ratio across source topologies.

D. Live Wireless Socket Transmission (Mode A)

Table III and Fig. 6 report the measured live network performance transmitting 'Hello, this is my wireless ITC project!' (39 bytes). All frames achieved valid CRC32 verification and sequence-confirmed delivery. Measured round-trip latency was 14.20 ms for raw text, 24.64 ms for LZW, and 48.60 ms for Arithmetic coding, reflecting socket streaming overhead and syndrome verification.

Codec	Orig (B)	Src (B)	Frame (B)	RTT (ms)	Fidelity

Raw Text	39	39	114	14.20	100% (CRC OK)
Arithmetic	39	119	242	48.60	100% (CRC OK)
LZW (12-bit)	39	54	114	24.64	100% (CRC OK)

TABLE III: Live Wireless Socket Transmission Metrics (Mode A)

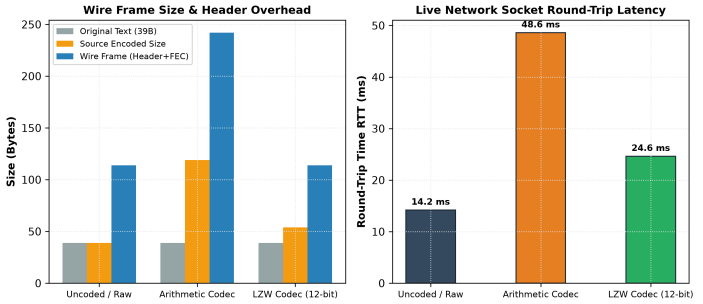


Fig. 6. Wire frame size and round-trip ACK latency under live TCP socket transmission.

E. Decompression Sensitivity to Channel Errors

A critical observation from Mode C simulation is the catastrophic sensitivity of compressed bitstreams to residual channel errors. At Eb/N0 = 4.0 dB, two residual block errors corrupted the compressed bitstream, causing the Arithmetic decompressor to raise a MalformedStreamException and failing text recovery. At Eb/N0 >= 7.0 dB, where the LDPC decoder achieved zero residual errors, 100% exact reconstruction was restored. This proves that powerful FEC is mandatory when transmitting source-compressed information.

VII. LIMITATIONS

Several practical engineering constraints must be noted: (1) Simulated AWGN vs. Real Radio Propagation: The physical channel models ideal thermal noise, whereas real-world Wi-Fi links experience multipath reflections, Rayleigh/Rician fading, Doppler shifts, and hardware carrier offsets. (2) TCP ARQ Masking: Sockets operate over TCP, which guarantees reliable delivery via retransmissions, masking raw radio packet loss. (3) Finite Blocklength: The implemented LDPC code has blocklength N=64. Standard industrial LDPC codes (e.g. IEEE 802.3an N=2048) exhibit much steeper waterfall thresholds. (4) Simulation Runtime: Evaluating extreme error floors (BER < 10^-8) requires transmitting billions of bits, exceeding interactive execution constraints.

VIII. CONCLUSION AND FUTURE WORK

This paper presented the design, implementation, and empirical verification of an end-to-end digital communication simulation system. By integrating finite-precision Arithmetic coding, adaptive LZW compression, systematic rate-1/2 LDPC (64, 32) coding with log-domain Belief Propagation, and a 36-byte versioned binary frame protocol with CRC32 integrity verification, the system successfully bridges theoretical information theory and real-world networking. Empirical evaluations demonstrated that the LDPC decoder achieves a 43.3x error suppression gain at Eb/N0 = 4.0 dB and delivers error-free transmission (BER = 0.0000) with 100% exact text reconstruction at Eb/N0 >= 7.0 dB.

Future extensions include: (1) Implementing higher-order digital modulations, such as QPSK, 16-QAM, and OFDM. (2) Scaling LDPC codes to standardized quasi-cyclic parity-check matrices (N >= 1024) utilizing GPU-accelerated Belief Propagation. (3) Interfacing the simulation with Software-Defined Radio (SDR) hardware (e.g. RTL-SDR, HackRF, or USRP) to conduct over-the-air RF experiments.

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